Case Study — Uplift Modeling for Premium Upsell at Leboncoin (Cars)

Pilot restricted to the **cars** category. Self-contained document with embedded figures and feature-consistent storytelling.

1) Business Problem

Premium options (boosts/visibility for listings) are a key monetization lever. Blanket discounts increase adoption but also give away margin to sellers who would have boosted anyway (cannibalization).

Core question: *Within cars, who should receive a discount, and at which level (5%, 10%, 20%), to maximize incremental profit rather than gross sales?*

2) Experimental Setup & Data Snapshot

- **Randomized experiment** with four arms: Control (0%), 5%, 10%, 20% discount.
- **Assignment** at the user level for causal identification of treatment effects.
- **Features (illustrative)** used in modeling:
    - *Behavior*: recency, purchase frequency, average spend
    - *Engagement*: ad views, messages sent, search activity
    - *Demographics*: geography, device, account age
    - *History*: prior discount redemption / responsiveness

Data snapshot (synthetic)

user_id	recency_days	freq_tx	avg_spend	ad_views	messages	search_actions
U001	12	8	€25	140	6	28
U002	45	2	€12	30	1	12
U003	5	15	€30	260	10	36
U004	100	1	€10	8	0	3
U005	22	3	€18	70	2	14

geo	device	acct_age_days	prior_discount_redemp	treatment	boost_purchased
Île-de-France	Mobile	420	0	0%	1
Occitanie	Desktop	120	1	10%	1
Auvergne-Rhône-Alpes	Mobile	980	0	20%	1
Hauts-de-France	Desktop	60	0	0%	0
PACA	Mobile	260	0	5%	0

Illustrative subset built only from the declared feature families above; split into two lines for readability.

3) Modeling Methodology

Uplift modeling predicts the *incremental* effect of treatment vs. control.

- **Approach**: T-learner with gradient boosted trees (LightGBM)
- Model C: $\hat{p}0(x) = P(Y = 1 \mid T = 0, x)$
- Model T: $\hat{p}1(x) = P(Y = 1 \mid T = 1, x)$
- **Uplift score**: $U(x) = \hat{p}1(x) - \hat{p}0(x)$

Training: 70% train / 30% validation (stratified by treatment), transforms for skewed features, probability calibration, early stopping on AUC; report uplift metrics (Qini, AUUC).

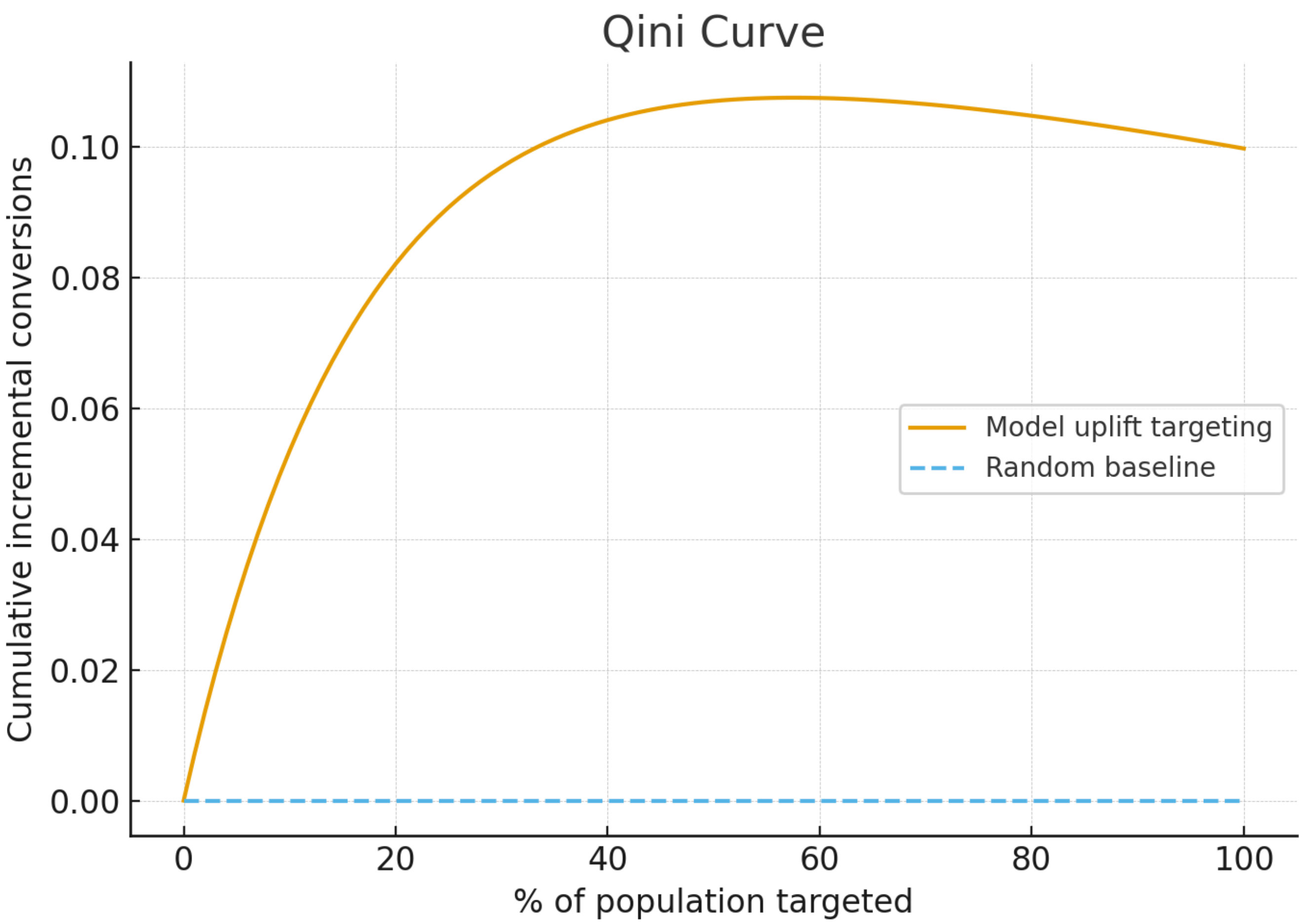
4) Evaluation & Results

Sellers are sorted by predicted uplift and binned into deciles; we compute incremental conversions using observed treatment vs. control outcomes in each bin.

- **Top decile**: ~+8.5% incremental uplift
- **Middle deciles**: ~0% uplift (neutral)
- **Bottom decile**: negative uplift (treatment harm)

Qini Curve

The curve below is a realistic shape: steep early gains (persuadables), then flattening as we target lower-uplift sellers. Dashed line is the random baseline.



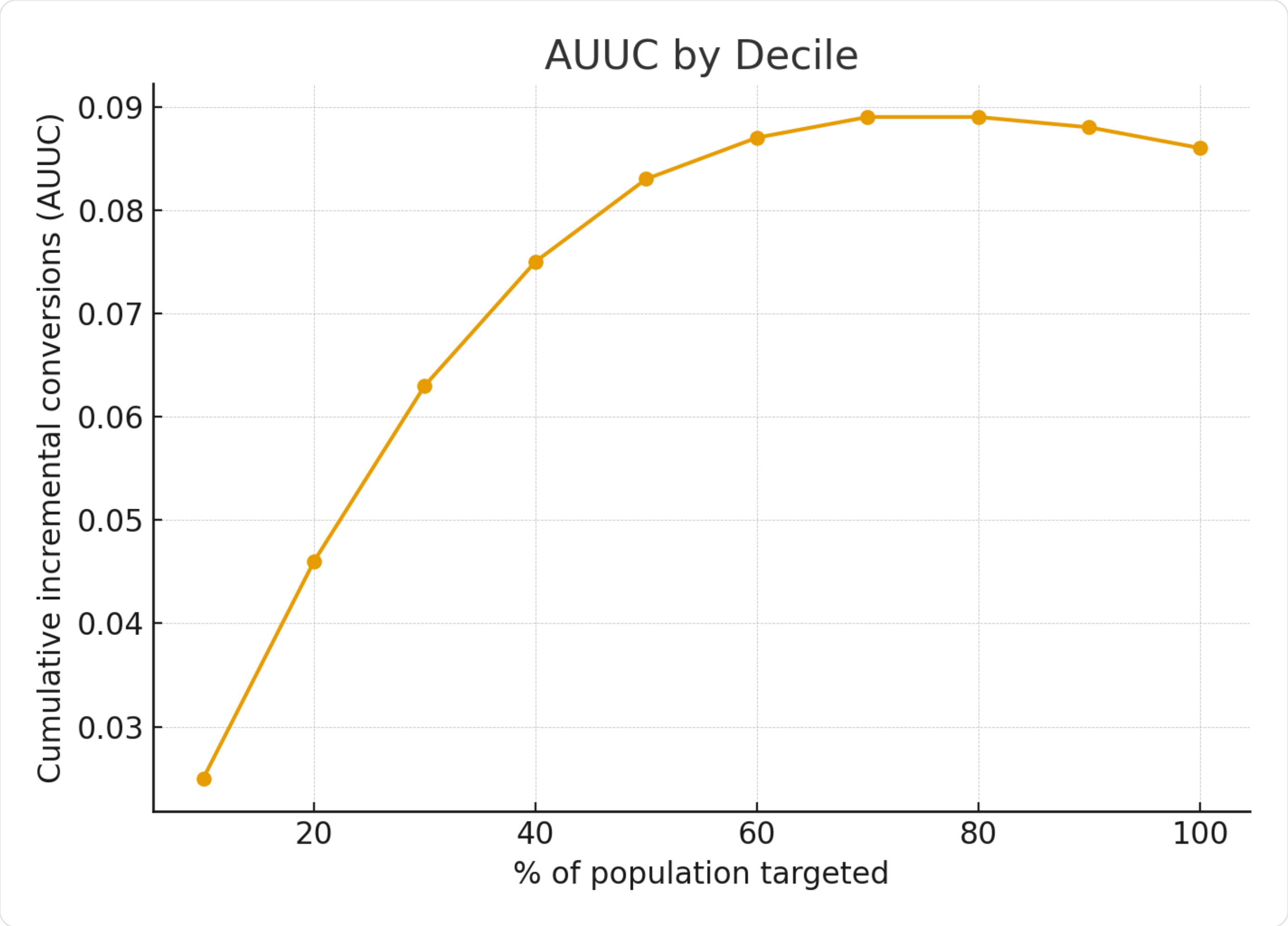
Cumulative incremental conversions vs. % targeted.

Headline metrics

- AUUC: +35% vs. random
- Qini coefficient: significantly > 0 (robust uplift signal)

AUUC by Decile

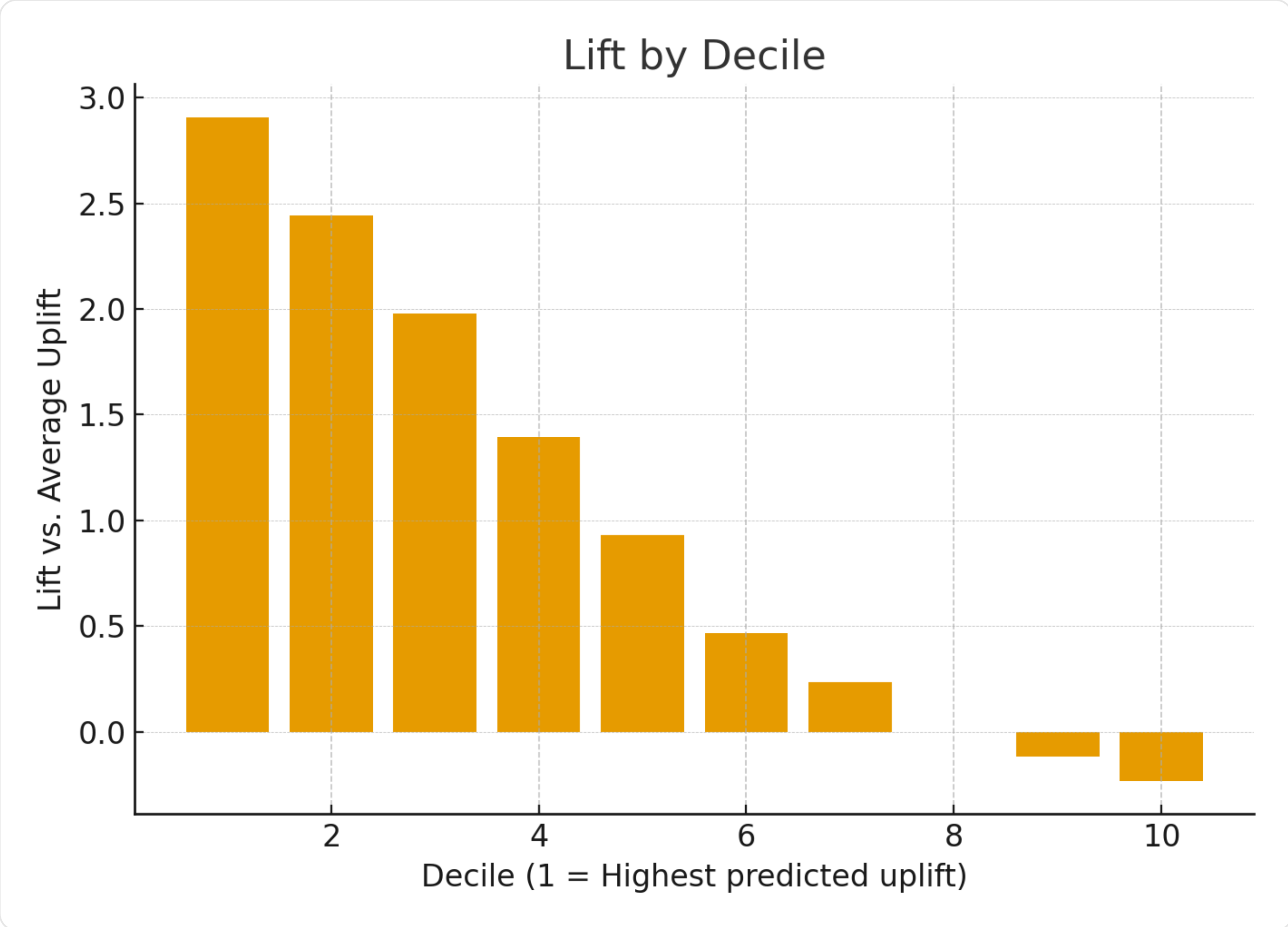
Cumulative incremental conversions as we expand targeting. Higher slope early = strong prioritization of persuadables.



Cumulative uplift across top deciles.

Lift by Decile

Decile-wise uplift vs. the overall average uplift (1 = average). Bars above 1 indicate segments worth prioritizing.



Relative lift by decile (1 = overall average).

5) Segmentation — Who Gets a Discount?

- **Persuadables (~25%)**: High positive uplift → *target*. Main budget focus.
- **Sure things (~20%)**: High baseline likelihood but low uplift → *exclude* to avoid margin leakage.
- **Lost causes (~45%)**: Very low conversion, near-zero uplift → *exclude*.
- **Do-not-disturb (~10%)**: Negative uplift → *explicitly exclude*.

6) Production Deployment

1. **Daily scoring** of active sellers → uplift score and recommended action
2. **Decision engine**
    - High uplift → 10% discount
    - Medium uplift → 5% discount
    - Others → no discount
3. **Guardrails**
    - Budget cap on discount cost
    - Exclusions: existing premium subscribers currently on an active boost
    - Max discount 20% to protect brand equity

4. **Monitoring**: Incremental profit dashboards; drift detection on uplift score distributions

7) Business Impact

- +5% **incremental boost adoption** in cars campaigns
- ~15% **reduction in discount leakage** (excluding sure things)
- +12% **ROI uplift**, combining incremental sales with avoided waste

8) Key Learnings

- Uplift models answer “who boosts *because* of the discount?” — not just “who boosts?”
- Decile-based evaluation (Qini curve) is intuitive for non-technical partners
- Named segments (Persuadables, Sure Things, Lost Causes, Do-not-disturb) simplify operations and comms
- Guardrails + monitoring are essential for safe, durable impact

9) Why This Uplift Model Is Credible (Cars, Feature-Consistent)

What the model is leveraging

- **Behavior**: Sellers with *recent platform activity* and *moderate past premium spend* show higher uplift than very inactive users or heavy habitual spenders.
- **Engagement**: *Ad views* and *messages* at moderate levels correlate with uplift (enough traction to benefit from a boost, but not so much that a boost is unnecessary).
- **Demographics**: *Account age* and *geography* contribute small but consistent signals (newer accounts in dense regions tend to be more uplift-sensitive).
- **History**: *Prior discount redemption* is a strong indicator of price sensitivity; light past responsiveness predicts positive incremental effect for small offers.

Typical persuadable seller profiles (built only from declared features)

- **Profile A — Engaged, value-conscious**: recent activity; moderate ad views/messages; low-to-moderate prior spend; few prior redemptions → responds to a 10% offer.
- **Profile B — Returning but selective**: medium purchase frequency of boosts; average spend in mid-range; has redeemed small discounts before → uplift positive for 5% offers.
- **Profile C — New-ish, actively searching**: younger account age; high search actions; modest engagement on current ads; minimal prior discount use → uplift emerges with a light nudge.

Why this supports the curves & ROI

- Early Qini gains reflect these persuadables: they have signal in engagement and history but don’t convert without a nudge.

- Filtering out sure things (heavy habitual spenders) and lost causes (very low engagement) reduces discount leakage.
- The model’s signals align with marketplace intuition while strictly using the documented feature families.